

WEEK 1 ASSIGNMENT

Cybersecurity Fundamentals

Foundations, Threats, Tools & the Role of a
Cybersecurity Analyst



Course	Cyber Security Fundamentals
Instructor	Talal Rehan
Institute	IT-SIMPLERA Institute
Module	Week 1 - Introduction to Cybersecurity

Five Real-World Cyberattacks & the CIA Triad

The CIA Triad — Confidentiality, Integrity, and Availability — is the standard lens for classifying the impact of a security incident. The five incidents below are among the most studied breaches in the industry, each illustrating a different way the triad can be broken.

Equifax Data Breach (2017)

Confidentiality

What happened: Attackers exploited an unpatched Apache Struts vulnerability in a consumer complaint web portal, gaining access to internal systems for over two months before detection.

Impact: Personal and financial records — names, Social Security numbers, birth dates, and addresses — of roughly 147 million people were exposed to unauthorized parties.

WannaCry Ransomware (2017)

Availability

What happened: A self-propagating worm exploited the Windows SMB vulnerability (EternalBlue) to spread across unpatched machines worldwide, encrypting files and demanding Bitcoin ransom.

Impact: Hospitals (including parts of the UK's NHS), manufacturers, and government agencies lost access to critical systems and records, halting operations in over 150 countries.

Stuxnet (2010)

Integrity

What happened: A highly sophisticated worm targeted Siemens industrial control systems (SCADA/PLCs) used in Iran's uranium enrichment facilities.

Impact: The malware secretly altered centrifuge rotor speeds while feeding operators falsified “normal” readings, causing physical equipment damage without triggering alarms.

Colonial Pipeline Ransomware Attack (2021)

Availability

What happened: Attackers used a single compromised VPN password (no multi-factor authentication) to gain network access, then deployed ransomware against IT systems.

Impact: The company proactively shut down pipeline operations supplying roughly 45% of fuel to the U.S.

East Coast, triggering regional shortages and panic buying for several days.

SolarWinds Supply-Chain Attack (2020)

Integrity & Confidentiality

What happened: Nation-state actors inserted malicious code (“Sunburst”) into a legitimate software update for the Orion network-management platform, distributed to thousands of customers.

Impact: Trojanized updates altered trusted software and opened a covert backdoor into numerous U.S. government agencies and Fortune 500 companies, exposing sensitive internal data.

2 Three Cybersecurity Tools

Security analysts rely on a mix of open-source and commercial tools to monitor networks, find weaknesses, and test defenses before attackers do. Three foundational tools are summarized below.

Tool	Category	What It Does	Typical Use Case
Wireshark	Network protocol analyzer	Captures live network traffic and lets analysts inspect individual packets down to the byte level, decoding hundreds of protocols.	Diagnosing network performance problems, spotting unusual traffic patterns, and investigating suspected intrusions.
Nmap	Network scanner	Discovers live hosts on a network, scans for open ports, and fingerprints the operating systems and services running on each device.	Building an asset inventory and identifying exposed or unnecessary services before attackers do.
Burp Suite	Web application security tester	Intercepts and manipulates traffic between a browser and a web application to surface flaws such as injection points and broken authentication.	Penetration testers auditing websites and APIs for exploitable vulnerabilities before release.

3 Secure Home Network Diagram

The diagram below shows a typical home network hardened with basic best practices: a single trusted entry point (the router/firewall), WPA3 encryption, and an isolated guest network kept separate from personal devices.

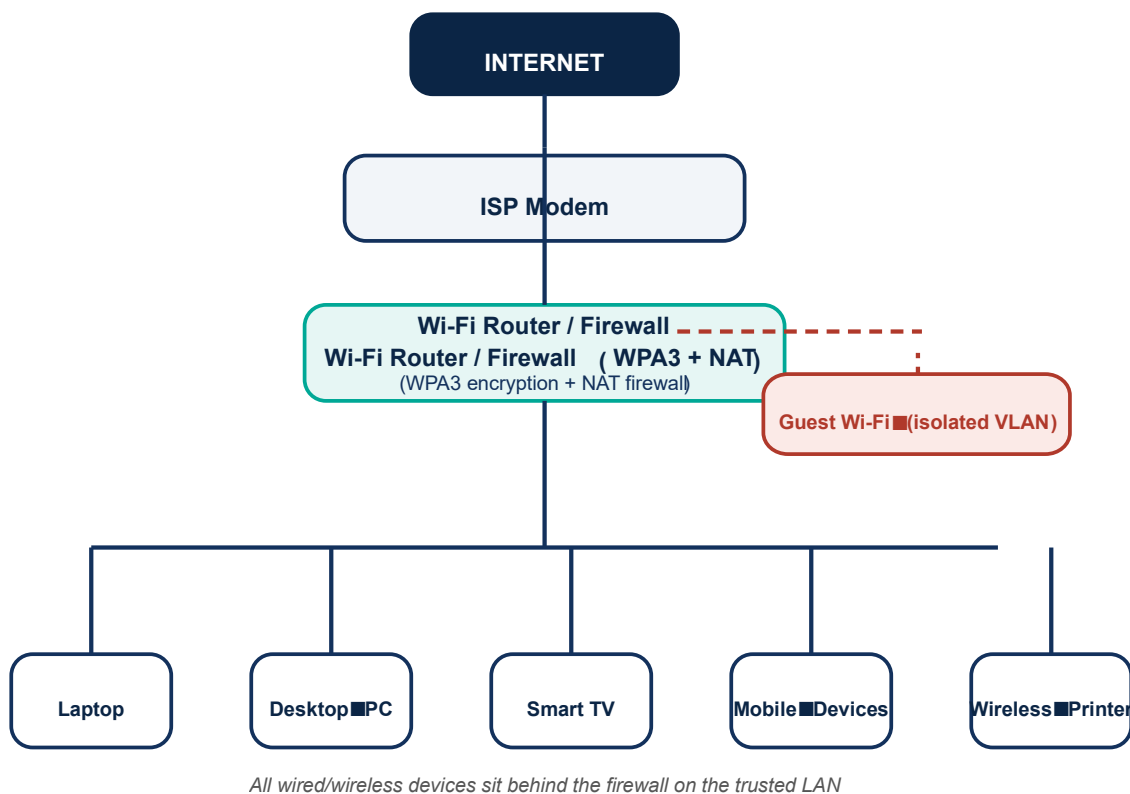


Figure 1. Secure home network topology, from ISP modem down to end devices.

Design principles applied:

- Router/Firewall as the single gateway: all inbound/outbound traffic passes through one hardened device performing NAT and stateful firewalling.
- WPA3 (or WPA2 minimum) encryption on the Wi-Fi network to prevent eavesdropping and unauthorized joins.
- Guest Wi-Fi on an isolated VLAN/SSID so visitors' devices cannot reach personal computers or files.
- Every device (laptop, desktop, smart TV, mobile, printer) sits behind the firewall rather than being exposed directly to the internet.

4

What Does a Cybersecurity Analyst Do Every Day?

A cybersecurity analyst's job is to keep an organization's systems, networks, and data safe from threats, and no two days look exactly the same. Still, most days revolve around a common rhythm of watching, investigating, fixing, and communicating.

The day typically starts with monitoring. Analysts review dashboards from tools such as SIEM platforms, firewalls, and antivirus software to see what happened overnight and what is happening right now. Most of these alerts turn out to be false positives, but each one has to be triaged quickly, because the one alert that gets ignored could be the real intrusion.

When something looks suspicious, the analyst moves into investigation. This means digging through logs from servers, applications, and network devices to reconstruct what actually happened: which account logged in, from where, and what it touched. Log analysis is detective work — piecing together small, disconnected clues into a clear timeline of an incident.

Alongside monitoring, analysts regularly run or review vulnerability assessments, scanning systems for outdated software, missing patches, and misconfigurations. Finding a weakness is only half the job; the analyst also has to prioritize it by risk and work with IT or development teams to get it fixed before an attacker finds it first.

Analysts also spend time on threat intelligence — reading up on newly discovered malware, vulnerabilities, and attacker techniques so the organization's defenses stay current against threats that did not even exist a few months earlier.

When a real incident is confirmed, the analyst shifts into incident response: helping contain the threat, stop it from spreading, and begin recovery, often alongside other members of the security team. Afterward comes reporting — writing up what happened, how it was handled, and what should change, in language that both technical colleagues and non-technical managers can understand.

Finally, a good analyst spends time on user awareness, helping the rest of the organization recognize phishing emails, choose strong passwords, and follow good cyber hygiene — because most breaches start with a person clicking the wrong link, not a broken firewall.

In short, a cybersecurity analyst's day blends technical skill with constant vigilance and clear communication: watching for trouble, chasing down what is real, fixing what is broken, and helping everyone else avoid becoming the next headline.